

Linear Regression Model for Council District VII

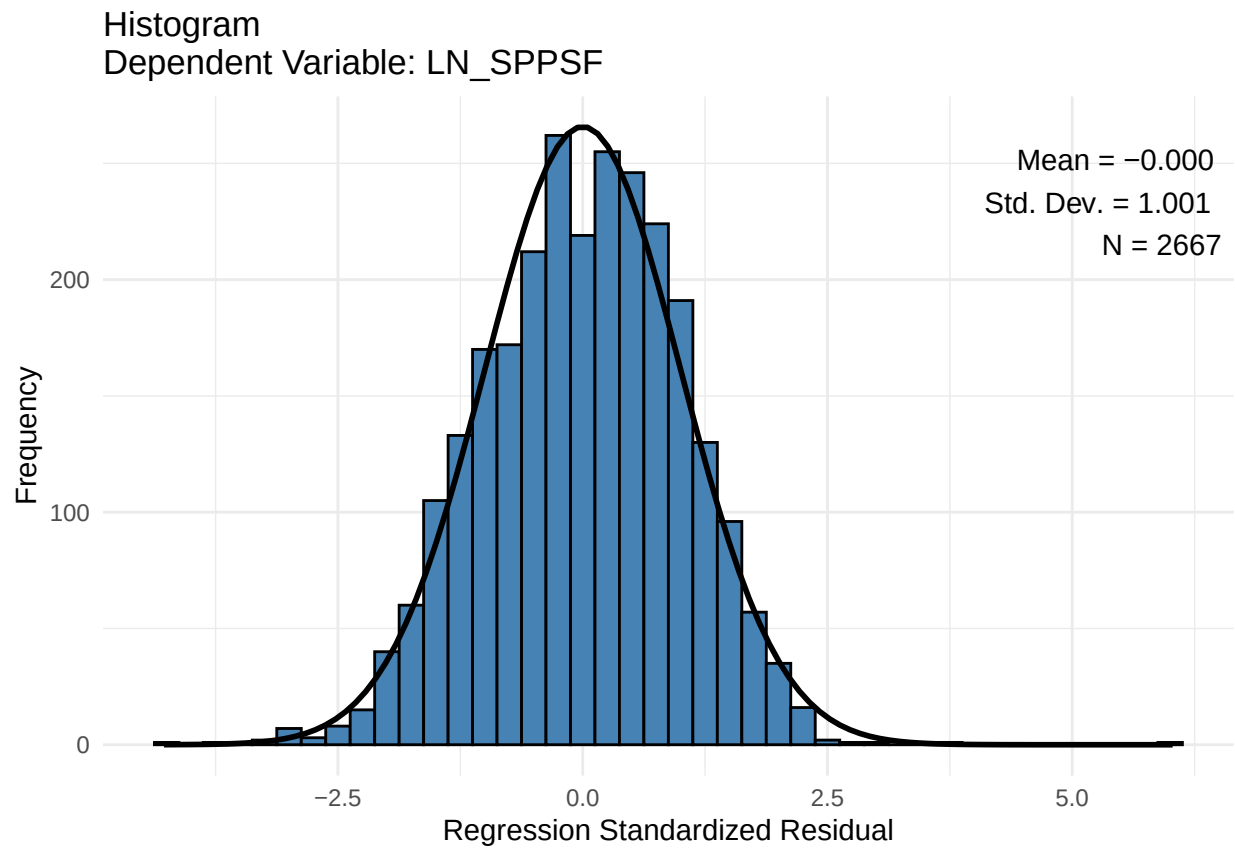
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Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

2 April 2026

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Model I



rate1	rate2	rate3	price_index
0.0162402	0.0109382	0	1.168671

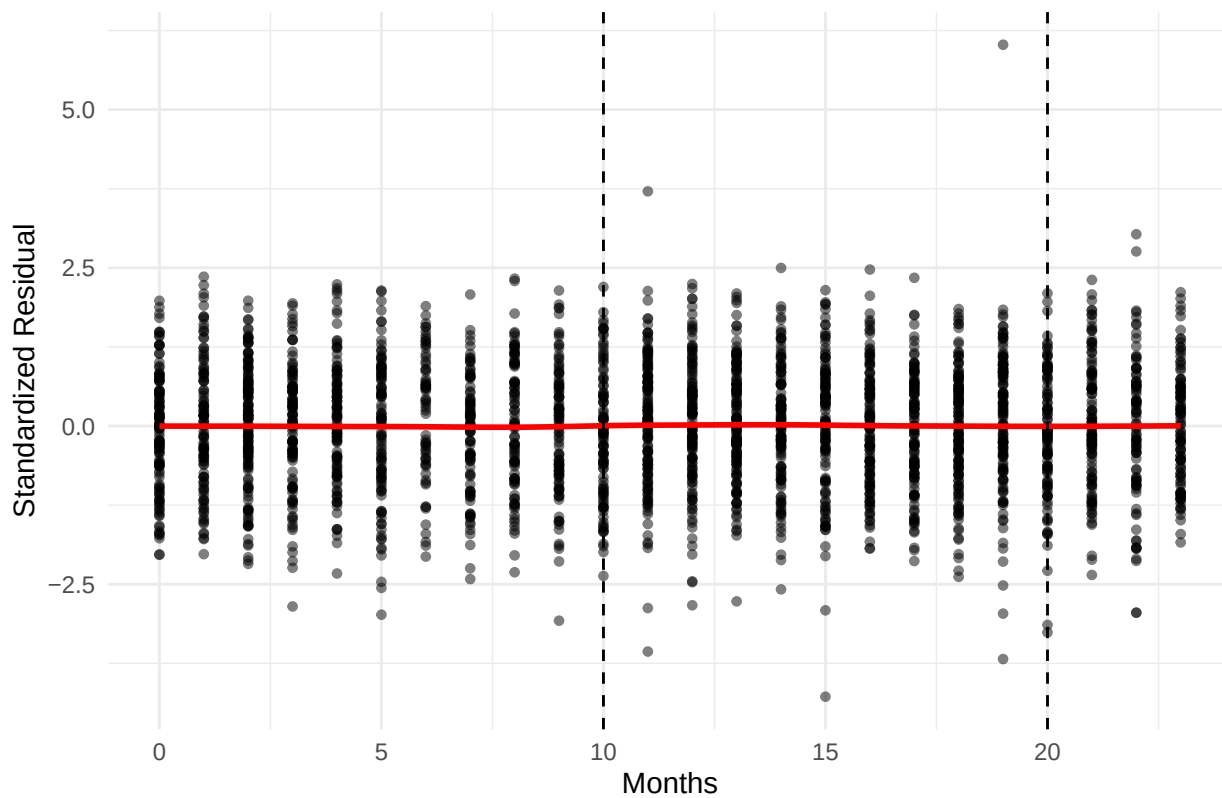
Table 2: District 7 LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2667	2646	21	0.4807	0.1973	0.1912	0.4788	0.3874	32.5144	0	3684.43	3813.981

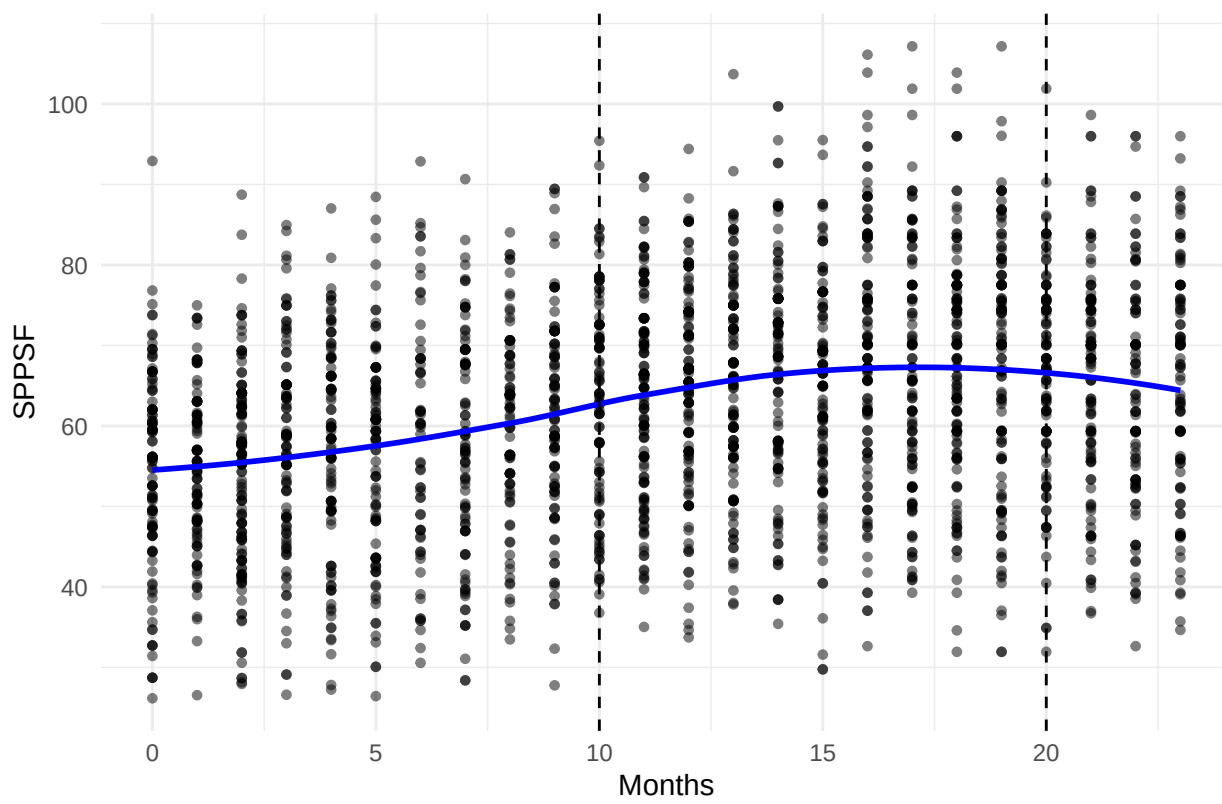
Table 3: District 7 LN_SPPSF model coefficients (95% CI)

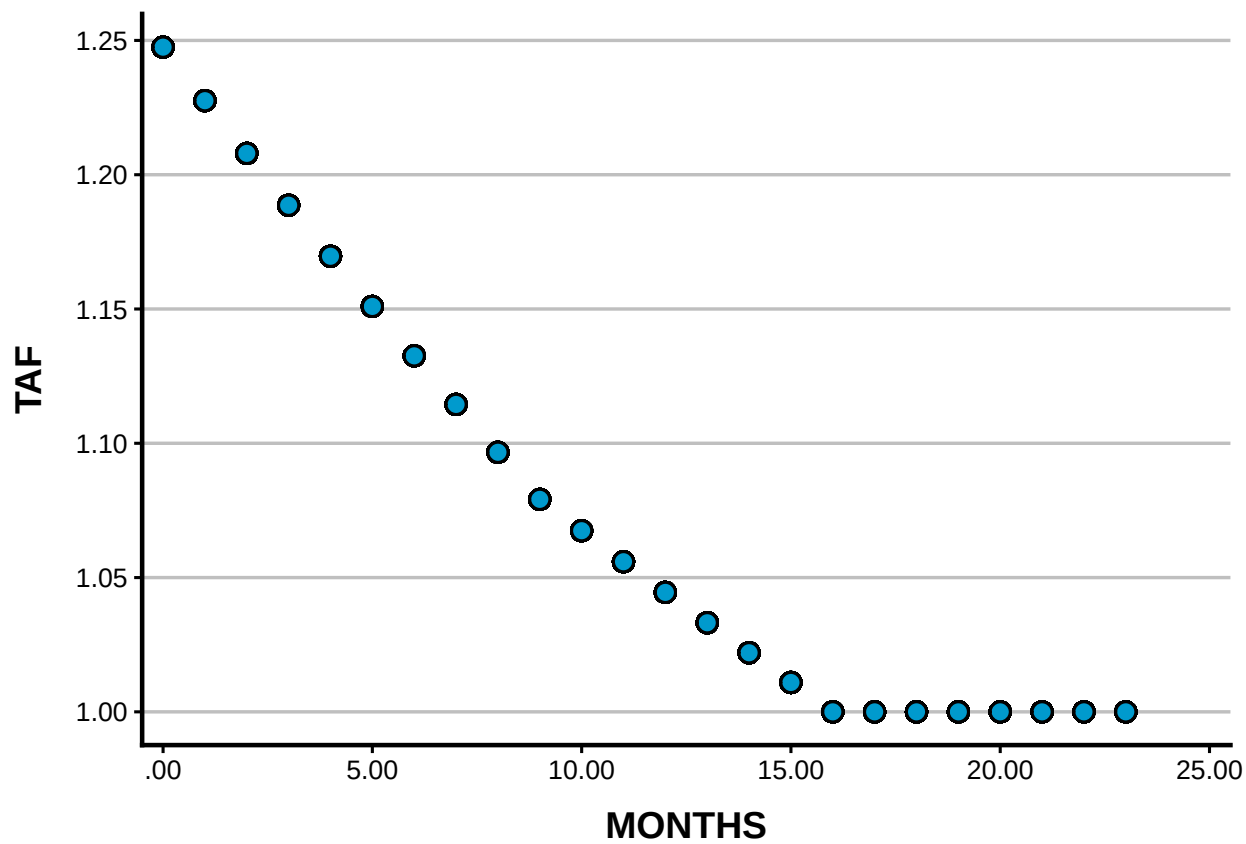
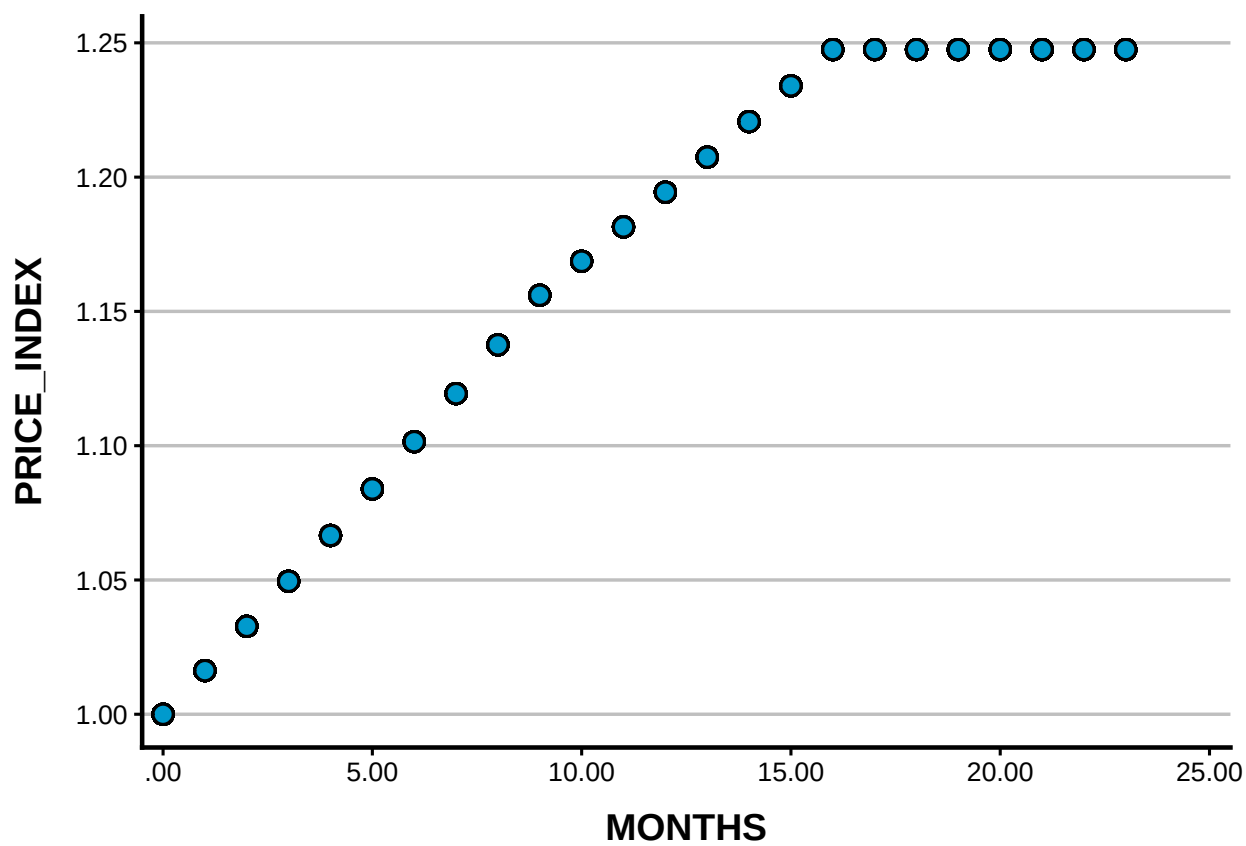
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.7940	92.2735	0.0411	0.0000	3.7133	3.8746
single_family1story	0.0669	2.1432	0.0312	0.0322	0.0057	0.1281
smallest	0.1789	5.4559	0.0328	0.0000	0.1146	0.2432
small	0.0998	4.1790	0.0239	0.0000	0.0530	0.1466
large	-0.0583	-1.8681	0.0312	0.0619	-0.1195	0.0029
fair	-0.1657	-7.1934	0.0230	0.0000	-0.2108	-0.1205
poor	-0.3141	-5.8079	0.0541	0.0000	-0.4201	-0.2080
qabove_avg	0.3249	2.6365	0.1232	0.0084	0.0833	0.5665
qbelow_avg	-0.0597	-2.6034	0.0229	0.0093	-0.1047	-0.0147
yb_pre_1929	-0.2473	-7.9388	0.0312	0.0000	-0.3084	-0.1862
yb_1946_to_1960	0.1016	4.6399	0.0219	0.0000	0.0586	0.1445
yb_post_1961	0.2018	2.9256	0.0690	0.0035	0.0666	0.3371
crawl	0.1515	1.9788	0.0766	0.0479	0.0014	0.3016
slab	-0.1948	-2.3066	0.0845	0.0212	-0.3604	-0.0292
ln_garagespaces	0.1805	7.5311	0.0240	0.0000	0.1335	0.2274
hot_water	0.1204	2.5339	0.0475	0.0113	0.0272	0.2135
elec_baseboard	-0.5878	-2.0978	0.2802	0.0360	-1.1372	-0.0384
wall	-0.1943	-1.8734	0.1037	0.0611	-0.3976	0.0091
central_air	0.1408	5.2951	0.0266	0.0000	0.0886	0.1929
months_1to9	0.0162	3.7303	0.0044	0.0002	0.0077	0.0248
months_10to16	0.0109	2.6667	0.0041	0.0077	0.0029	0.0190

Residuals for LN_SPPSF Model



Months vs Predicted SPPSF – LN_SPPSF Model





Model II

Histogram Dependent Variable: LN_PRICE (No ECFs)

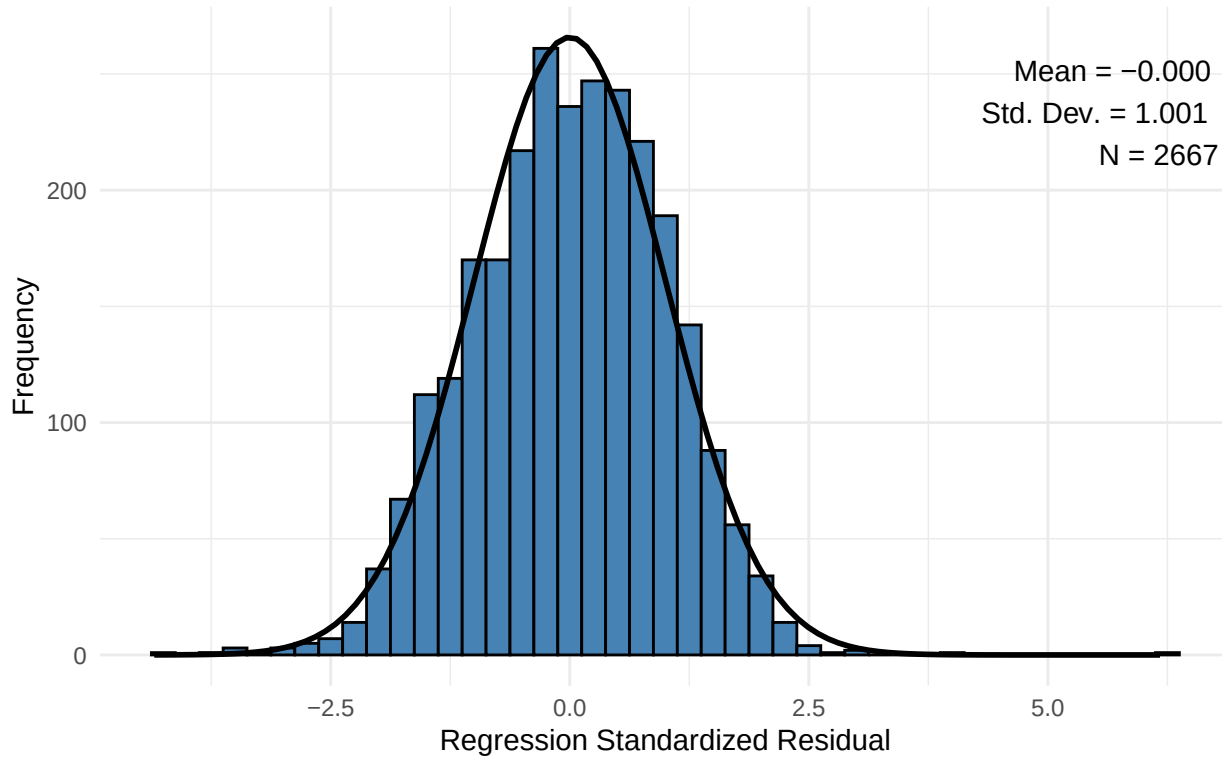


Table 4: District 7 LN_PRICE (No ECFs) model fit statistics

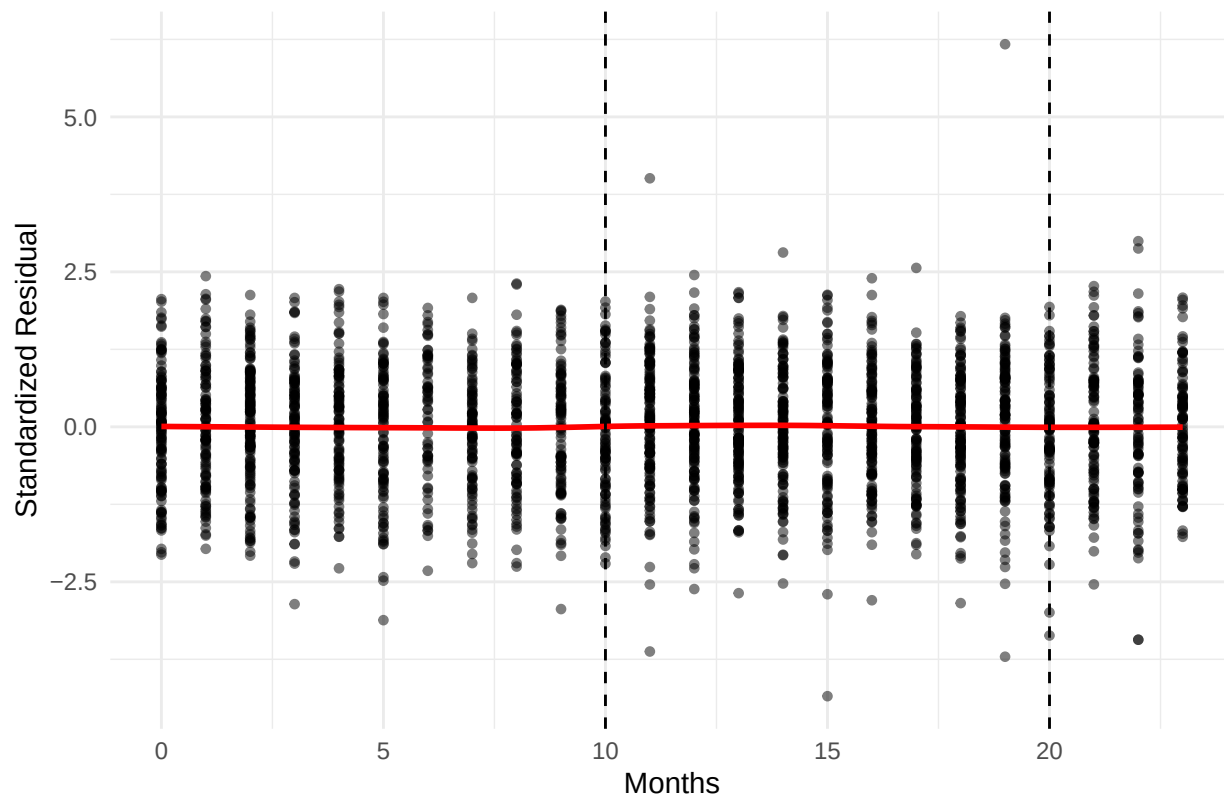
n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2667	2646	21	0.4755	0.2501	0.2445	0.4736	0.3822	44.1337	0	3626.189	3755.741

Table 5: District 7 LN_PRICE (No ECFs) model coefficients (95% CI)

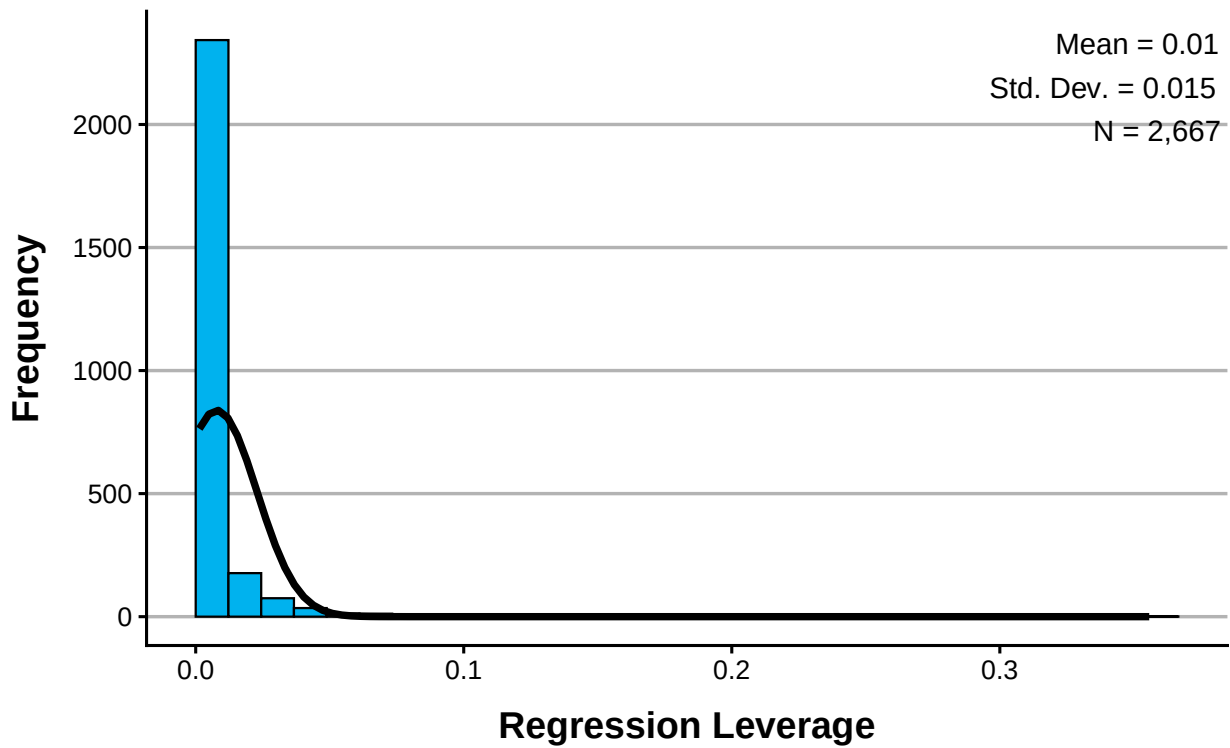
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.8117	275.3186	0.0393	0.0000	10.7347	10.8887
ln_landratio	0.3893	7.1400	0.0545	0.0000	0.2824	0.4962
ln_base_bldgsize_ratio	0.5990	10.4847	0.0571	0.0000	0.4870	0.7110
single_family1story	0.0652	2.0566	0.0317	0.0398	0.0030	0.1273
two_plus_baths	0.2331	2.6959	0.0865	0.0071	0.0636	0.4027
fair	-0.1641	-7.2007	0.0228	0.0000	-0.2088	-0.1194
poor	-0.2954	-5.5249	0.0535	0.0000	-0.4002	-0.1906
qabove_avg	0.2889	2.1998	0.1313	0.0279	0.0314	0.5465
qbelow_avg	-0.0751	-3.2504	0.0231	0.0012	-0.1205	-0.0298
yb_pre_1929	-0.2157	-7.0025	0.0308	0.0000	-0.2761	-0.1553
yb_1946_to_1960	0.0738	3.3709	0.0219	0.0008	0.0309	0.1167
yb_post_1961	0.2243	3.3054	0.0679	0.0010	0.0912	0.3573
crawl	0.1438	1.9088	0.0754	0.0564	-0.0039	0.2916
slab	-0.2332	-2.7837	0.0838	0.0054	-0.3974	-0.0689
ln_garagespaces	0.1590	6.6429	0.0239	0.0000	0.1121	0.2060
hot_water	0.1439	2.9446	0.0489	0.0033	0.0481	0.2398
elec_baseboard	-0.6149	-2.2134	0.2778	0.0270	-1.1597	-0.0702

term	estimate	statistic	std_error	p_value	conf_low	conf_high
wall	-0.2067	-2.0167	0.1025	0.0438	-0.4077	-0.0057
central_air	0.1352	5.1316	0.0263	0.0000	0.0835	0.1869
months_1to9	0.0151	3.5113	0.0043	0.0005	0.0067	0.0235
months_10to16	0.0126	3.1115	0.0041	0.0019	0.0047	0.0206

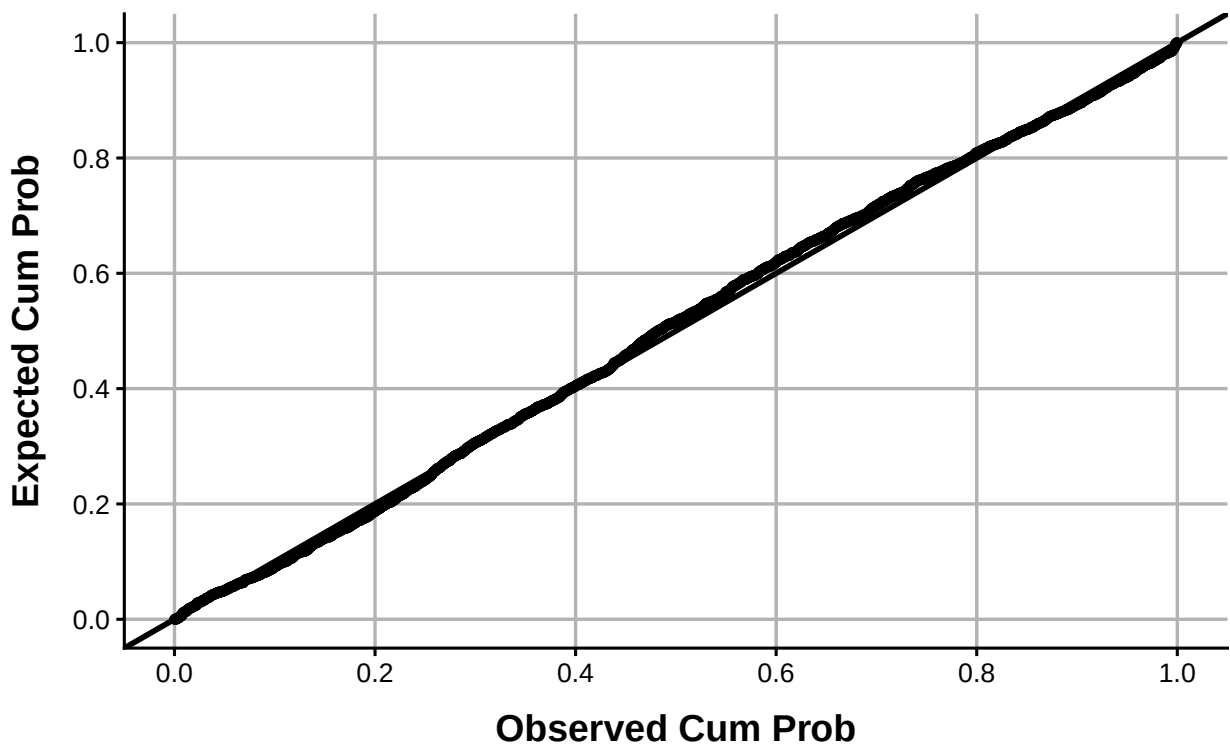
Residuals for LN_PRICE (No ECFs) Model



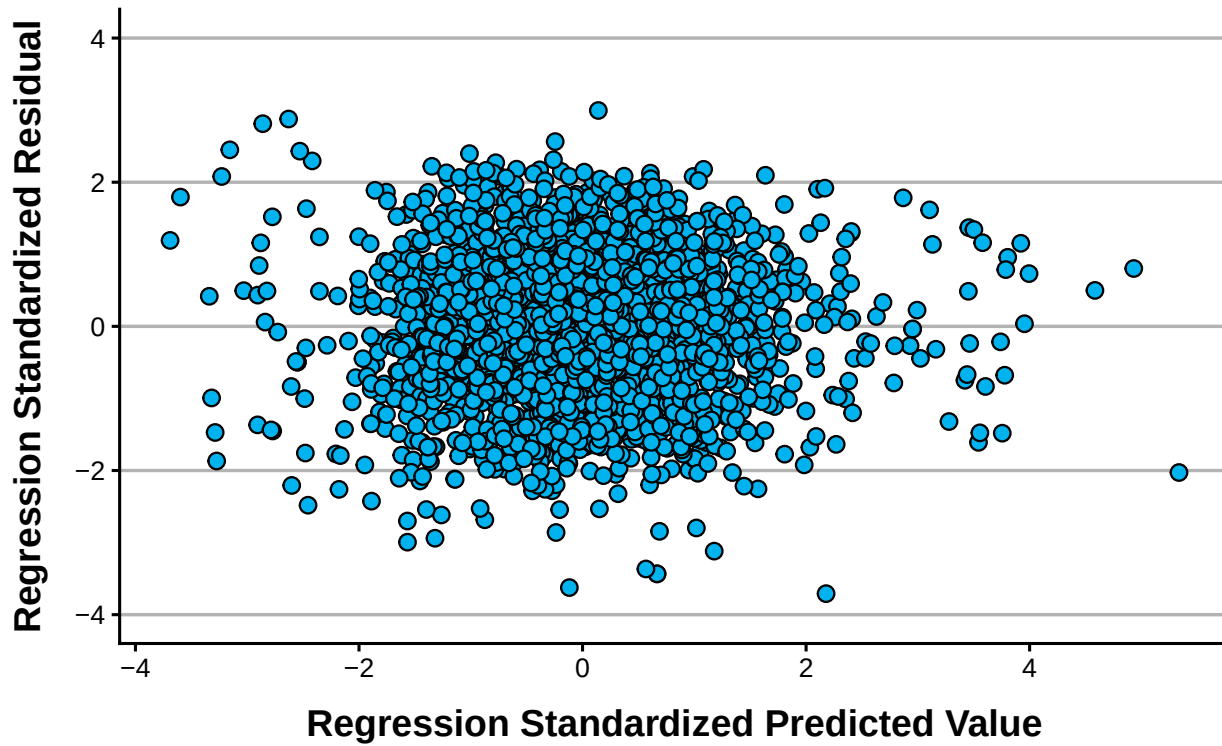
Histogram
Dependent Variable: LN_PRICE



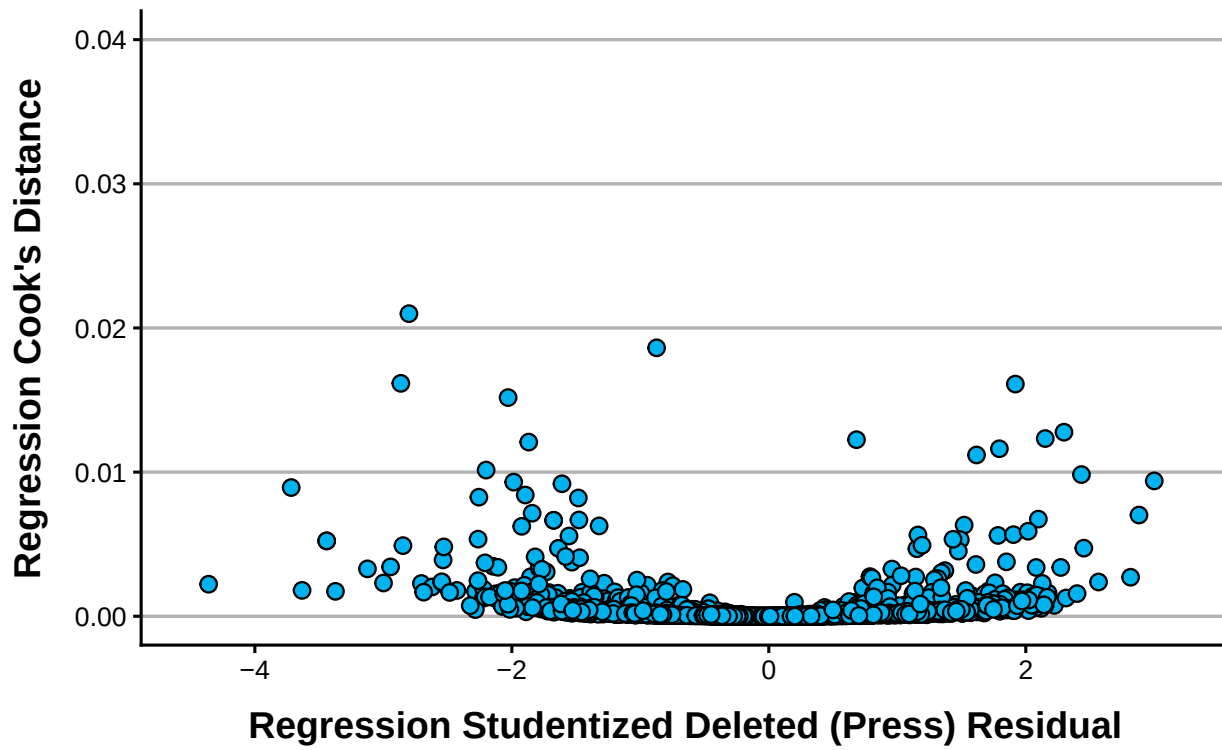
Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE

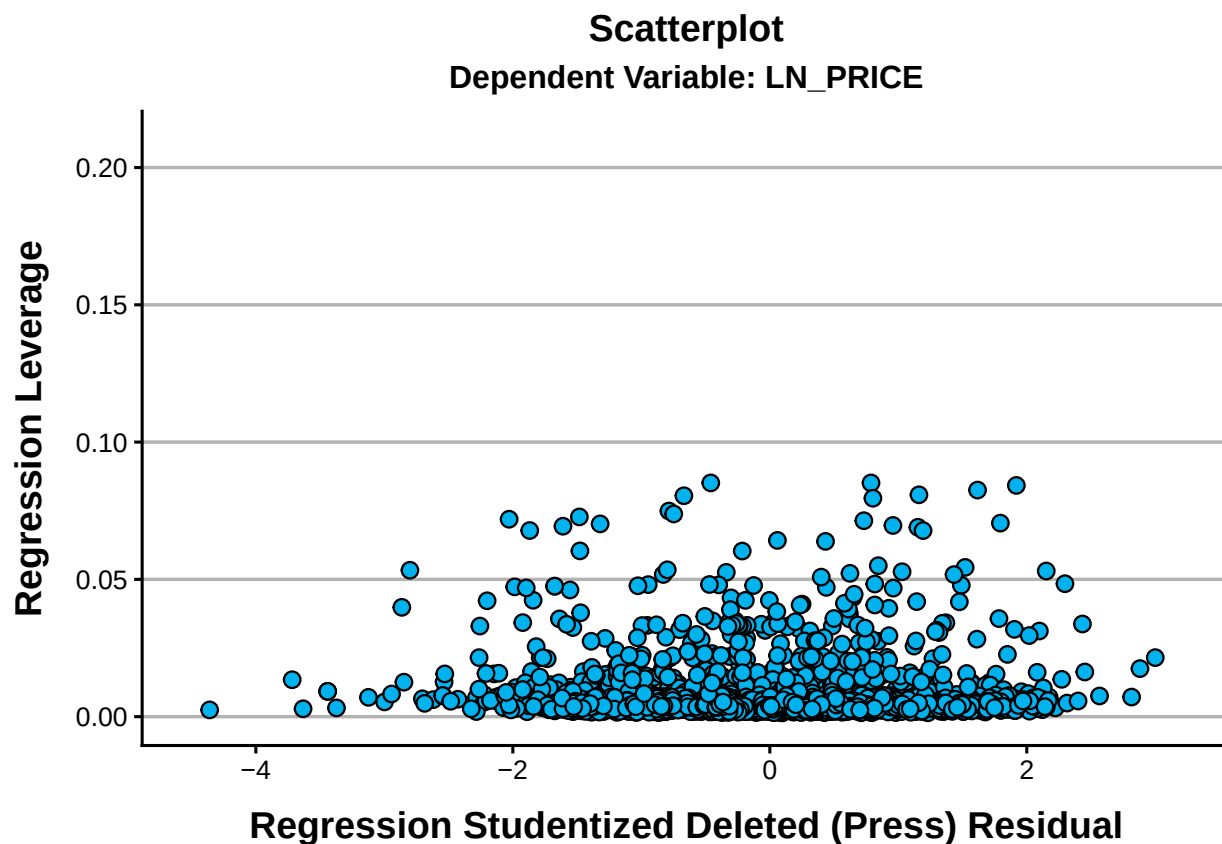


Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE





Model III

Initial Input

Table 6: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2667	0.9838809	1.102869	0.9120536	1.209215	38.96881	-0.7429556

Table 7: District 7 LN_PRICE model fit statistics

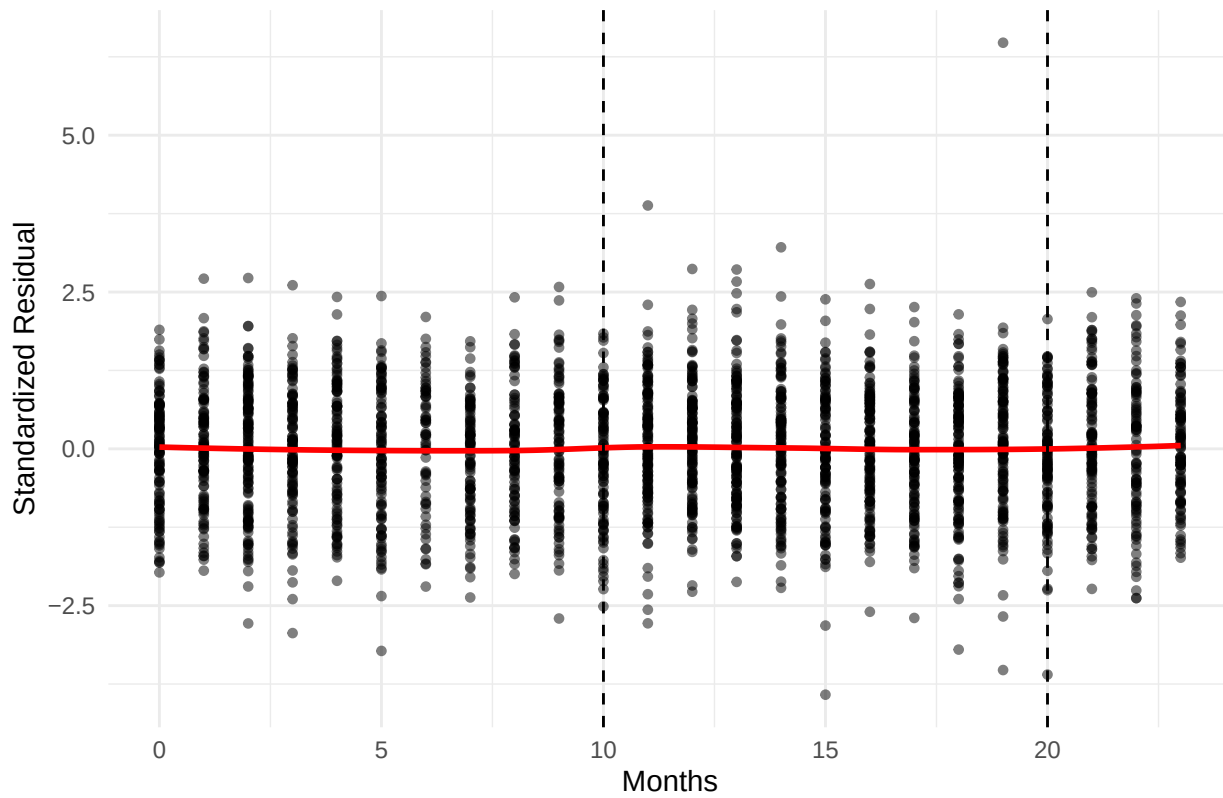
n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2667	2630	37	0.4443	0.3492	0.3403	0.4412	0.3573	39.2018	0	3280.273	3504.044

Table 8: District 7 LN_PRICE model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.3037	246.9065	0.0458	0.0000	11.2140	11.3935
ln_landratio	0.1717	3.0185	0.0569	0.0026	0.0602	0.2833
ln_base_bldgsize_ratio	0.5402	10.9573	0.0493	0.0000	0.4436	0.6369
fair	-0.1456	-6.7595	0.0215	0.0000	-0.1878	-0.1033
poor	-0.2821	-5.6031	0.0504	0.0000	-0.3809	-0.1834
qbelow_avg	-0.0741	-3.2509	0.0228	0.0012	-0.1188	-0.0294

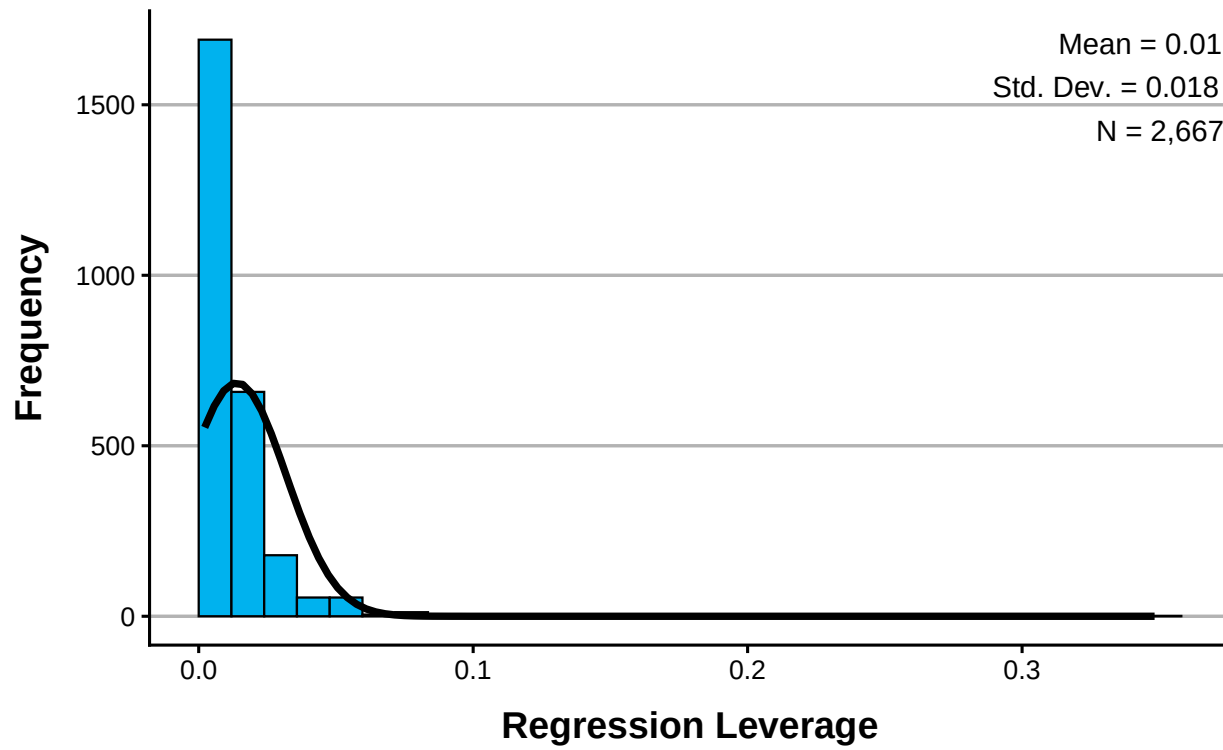
term	estimate	statistic	std_error	p_value	conf_low	conf_high
yb_pre_1929	-0.0654	-2.0873	0.0313	0.0370	-0.1268	-0.0040
yb_1946_to_1960	0.0396	1.8419	0.0215	0.0656	-0.0026	0.0817
yb_post_1961	0.1361	2.1670	0.0628	0.0303	0.0129	0.2593
crawl	0.1450	2.0493	0.0707	0.0405	0.0063	0.2837
ln_garagespaces	0.1260	5.5465	0.0227	0.0000	0.0814	0.1705
fireplace	0.0737	3.0410	0.0242	0.0024	0.0262	0.1212
elec_baseboard	-0.5383	-2.0728	0.2597	0.0383	-1.0476	-0.0291
wall	-0.2604	-2.7349	0.0952	0.0063	-0.4471	-0.0737
central_air	0.0905	3.6669	0.0247	0.0003	0.0421	0.1389
ecf_7R703	-0.6053	-8.6935	0.0696	0.0000	-0.7418	-0.4688
ecf_7R704	-0.4069	-10.4313	0.0390	0.0000	-0.4834	-0.3304
ecf_7R705	-0.4200	-12.3903	0.0339	0.0000	-0.4864	-0.3535
ecf_7R706	-0.4056	-5.1231	0.0792	0.0000	-0.5609	-0.2504
ecf_7R707	-0.4632	-5.4127	0.0856	0.0000	-0.6311	-0.2954
ecf_7R708	-0.4994	-9.8235	0.0508	0.0000	-0.5990	-0.3997
ecf_7R709	-0.4797	-9.8150	0.0489	0.0000	-0.5756	-0.3839
ecf_7R710	-0.2759	-5.3151	0.0519	0.0000	-0.3776	-0.1741
ecf_7R711	-0.5417	-11.1196	0.0487	0.0000	-0.6372	-0.4462
ecf_7R712	-0.4088	-5.8541	0.0698	0.0000	-0.5457	-0.2718
ecf_7R713	-0.4613	-7.4506	0.0619	0.0000	-0.5827	-0.3399
ecf_7R714	-0.7285	-3.5932	0.2027	0.0003	-1.1260	-0.3309
ecf_7R715	-0.6810	-6.8493	0.0994	0.0000	-0.8760	-0.4861
ecf_7R716	-0.9733	-14.8561	0.0655	0.0000	-1.1018	-0.8448
ecf_7R717	-0.8882	-13.6648	0.0650	0.0000	-1.0157	-0.7608
ecf_7R718	-0.7182	-14.1925	0.0506	0.0000	-0.8174	-0.6190
ecf_7R719	-0.3403	-8.7336	0.0390	0.0000	-0.4167	-0.2639
ecf_7R720	-0.7863	-6.1279	0.1283	0.0000	-1.0379	-0.5347
ecf_7R722	-0.5709	-5.4198	0.1053	0.0000	-0.7775	-0.3644
ecf_7R724	-0.7913	-9.9724	0.0793	0.0000	-0.9469	-0.6357
months_1to9	0.0154	3.8660	0.0040	0.0001	0.0076	0.0232
months_10to16	0.0151	3.9550	0.0038	0.0001	0.0076	0.0225

Residuals for LN_PRICE Model

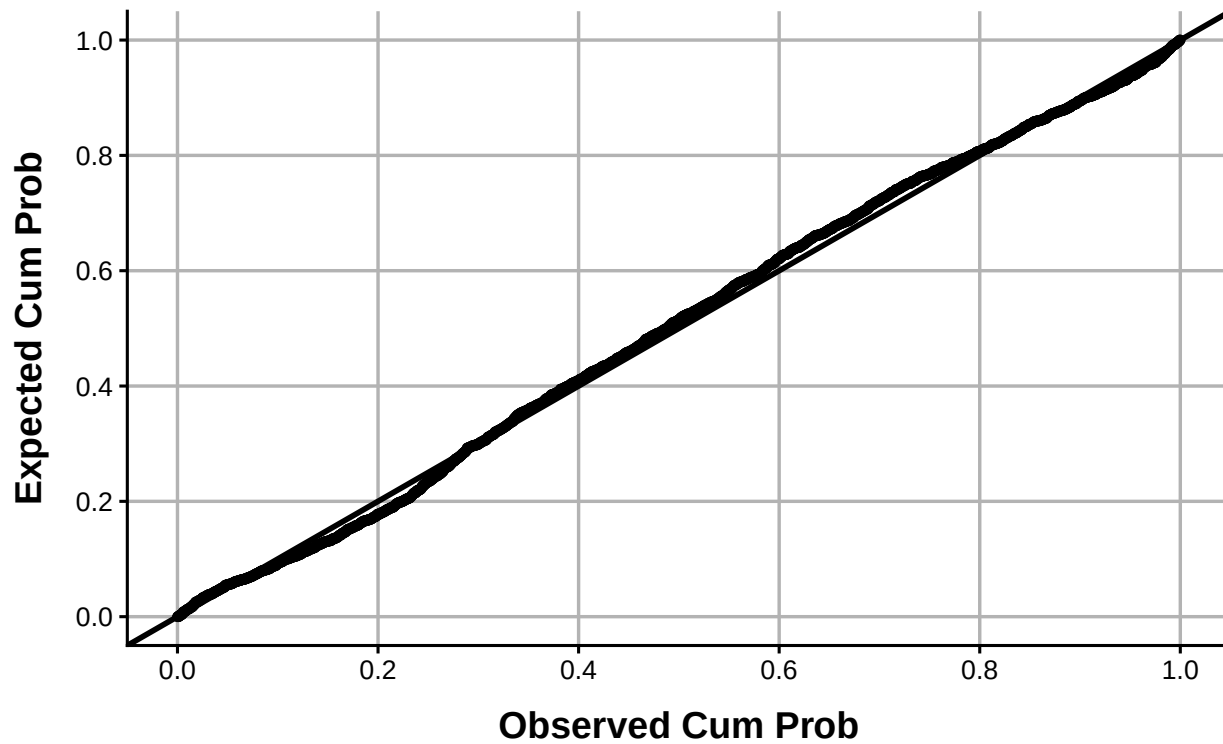


Histogram

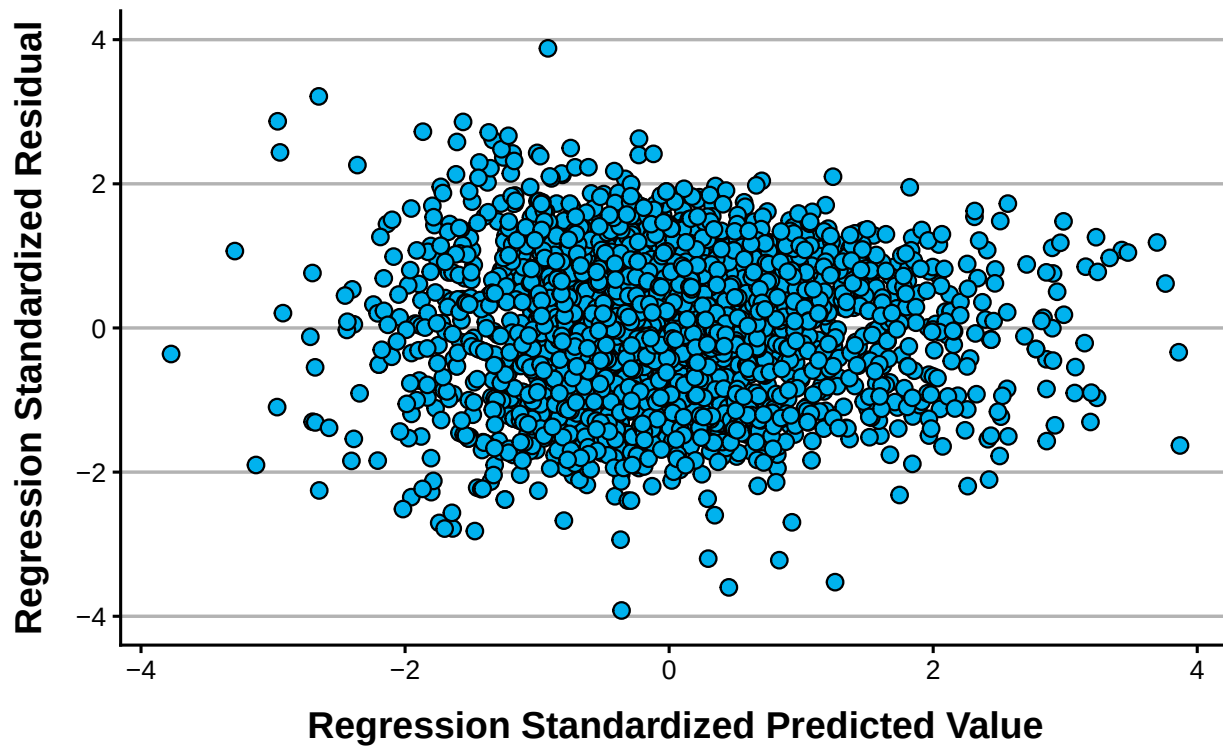
Dependent Variable: LN_PRICE

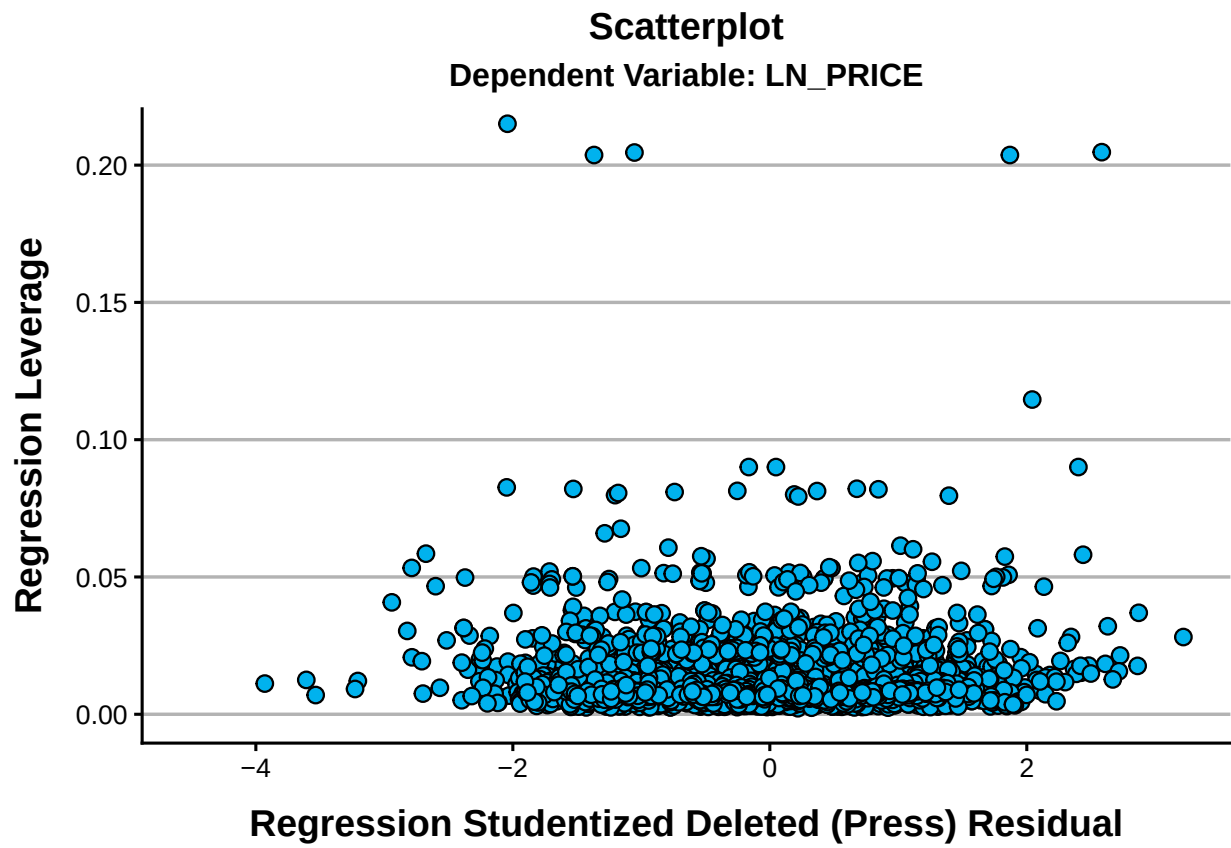
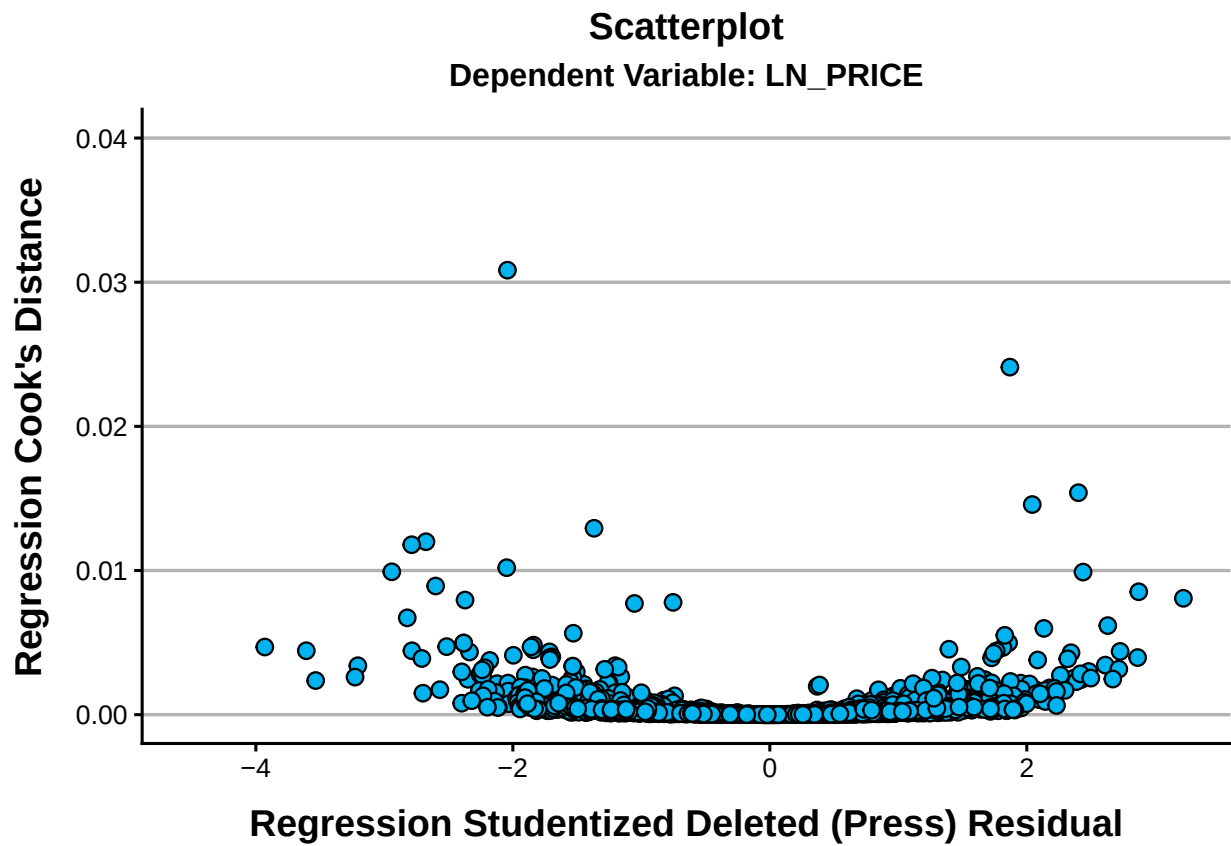


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE





Filtered Input

rate1_re	rate2_re	rate3_re	price_index_re
NA	NA	NA	NA

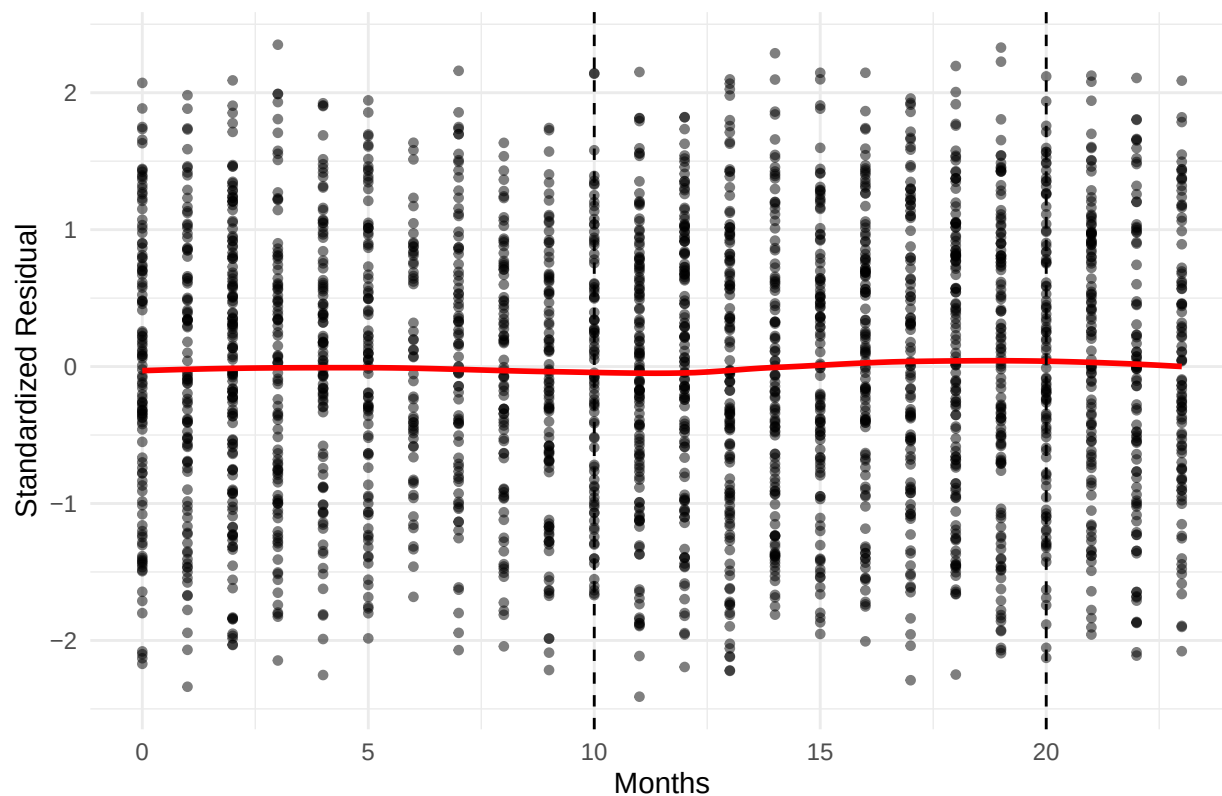
Table 10: District 7 LN_PRICE (Inliers) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2490	2454	38	0.3875	0.4098	0.4013	0.3847	0.3198	48.6757	0	2383.42	2598.762

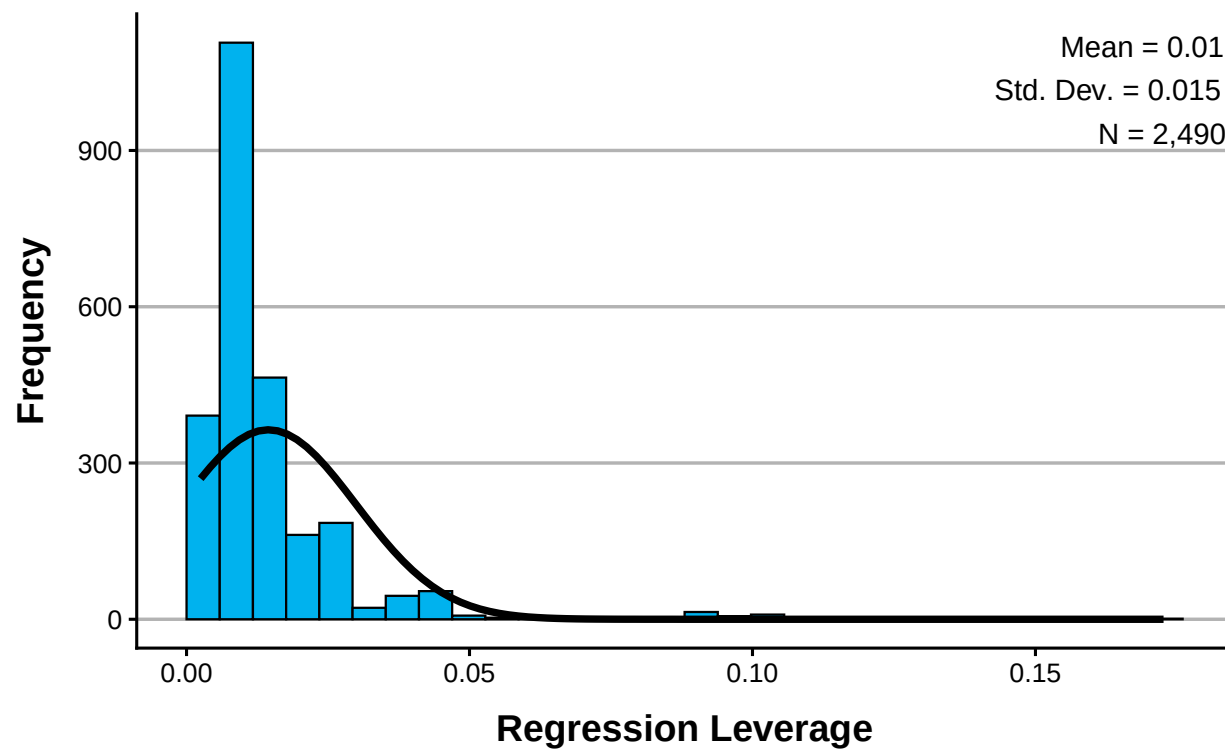
Table 11: District 7 LN_PRICE (Inliers) model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.2579	254.6984	0.0442	0.0000	11.1712	11.3445
ln_landratio	0.2479	4.7113	0.0526	0.0000	0.1447	0.3510
ln_base_bldgsize_ratio	0.5044	8.7702	0.0575	0.0000	0.3916	0.6172
single_family1story	0.0471	1.7017	0.0277	0.0889	-0.0072	0.1013
smallest	-0.0569	-1.9329	0.0295	0.0534	-0.1147	0.0008
fair	-0.1249	-6.3685	0.0196	0.0000	-0.1633	-0.0864
poor	-0.2807	-5.4172	0.0518	0.0000	-0.3823	-0.1791
qbelow_avg	-0.0792	-3.8593	0.0205	0.0001	-0.1195	-0.0390
yb_1946_to_1960	0.0441	2.3951	0.0184	0.0167	0.0080	0.0801
yb_post_1961	0.2138	3.6779	0.0581	0.0002	0.0998	0.3277
crawl	0.2065	2.7663	0.0747	0.0057	0.0601	0.3529
ln_garagespaces	0.1348	6.5858	0.0205	0.0000	0.0946	0.1749
fireplace	0.0744	3.3668	0.0221	0.0008	0.0311	0.1177
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	-0.3161	-2.7733	0.1140	0.0056	-0.5396	-0.0926
central_air	0.0848	3.8712	0.0219	0.0001	0.0419	0.1278
ecf_7R703	-0.6271	-9.8402	0.0637	0.0000	-0.7521	-0.5021
ecf_7R704	-0.4080	-11.8317	0.0345	0.0000	-0.4756	-0.3403
ecf_7R705	-0.4113	-13.7537	0.0299	0.0000	-0.4700	-0.3527
ecf_7R706	-0.3889	-4.9962	0.0778	0.0000	-0.5416	-0.2363
ecf_7R707	-0.4173	-5.0680	0.0823	0.0000	-0.5787	-0.2558
ecf_7R708	-0.4950	-10.9673	0.0451	0.0000	-0.5835	-0.4065
ecf_7R709	-0.4677	-10.7924	0.0433	0.0000	-0.5526	-0.3827
ecf_7R710	-0.2895	-6.3026	0.0459	0.0000	-0.3796	-0.1994
ecf_7R711	-0.5180	-12.0371	0.0430	0.0000	-0.6024	-0.4337
ecf_7R712	-0.3508	-5.5080	0.0637	0.0000	-0.4757	-0.2259
ecf_7R713	-0.4631	-8.1601	0.0568	0.0000	-0.5744	-0.3518
ecf_7R714	NA	NA	NA	NA	NA	NA
ecf_7R715	-0.6163	-5.3113	0.1160	0.0000	-0.8438	-0.3888
ecf_7R716	-0.9277	-15.0631	0.0616	0.0000	-1.0485	-0.8070
ecf_7R717	-0.9747	-16.3133	0.0597	0.0000	-1.0919	-0.8575
ecf_7R718	-0.7425	-16.3252	0.0455	0.0000	-0.8317	-0.6533
ecf_7R719	-0.3279	-9.5122	0.0345	0.0000	-0.3955	-0.2603
ecf_7R720	-0.7197	-4.7884	0.1503	0.0000	-1.0144	-0.4250
ecf_7R722	-0.4936	-3.8677	0.1276	0.0001	-0.7439	-0.2434
ecf_7R724	-0.8466	-10.4848	0.0807	0.0000	-1.0049	-0.6882
months_1to9	0.0135	3.7263	0.0036	0.0002	0.0064	0.0206
months_10to16	0.0165	4.7845	0.0034	0.0000	0.0097	0.0232

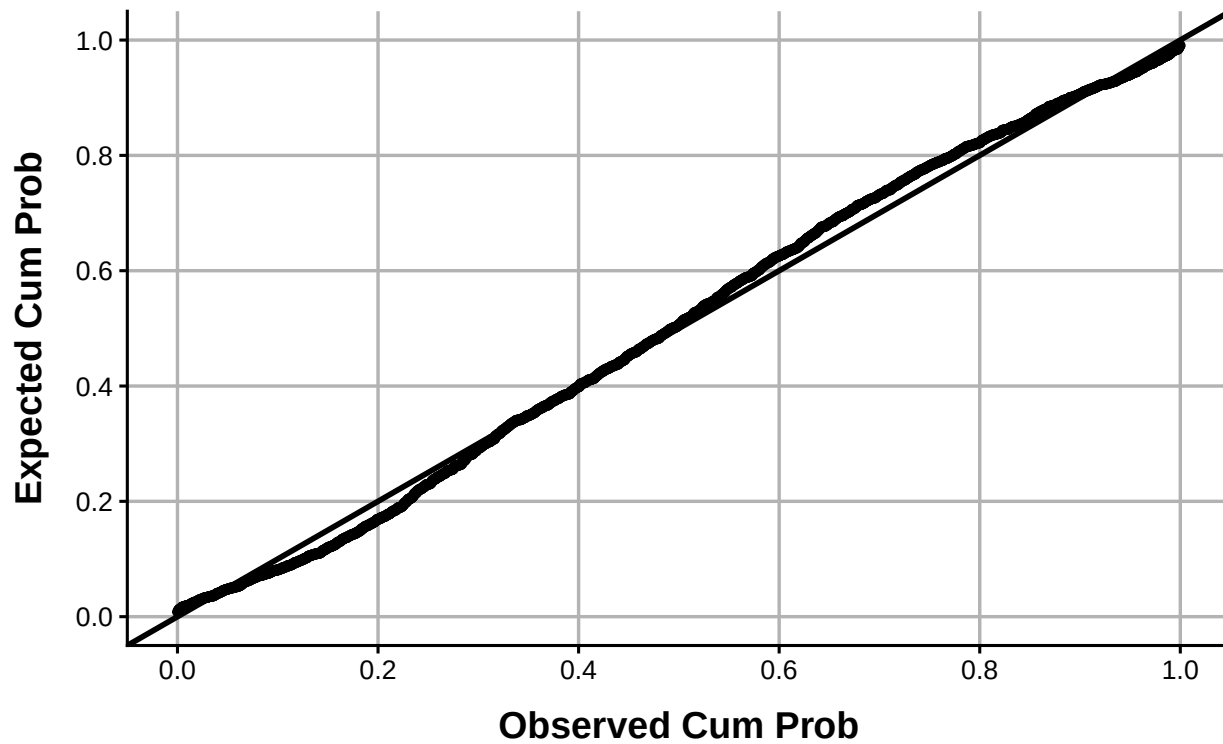
Residuals for LN_PRICE (Inliers) Model



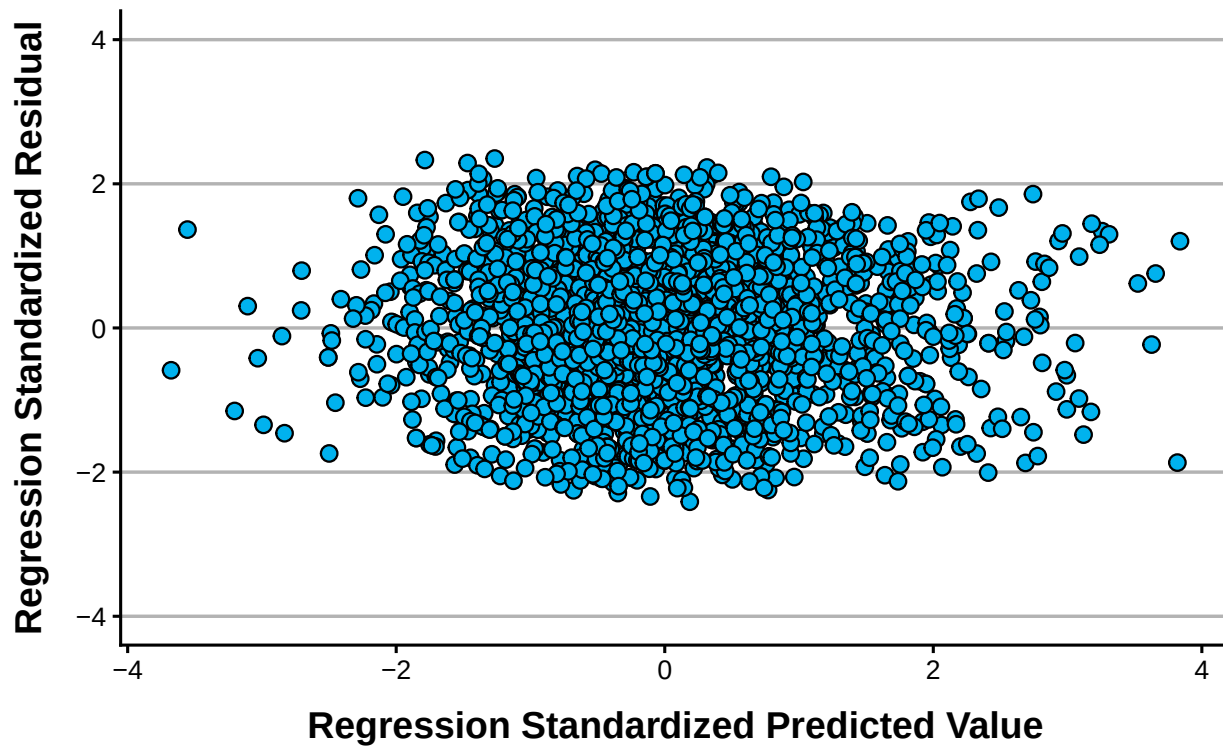
Histogram
Dependent Variable: LN\ PRICE



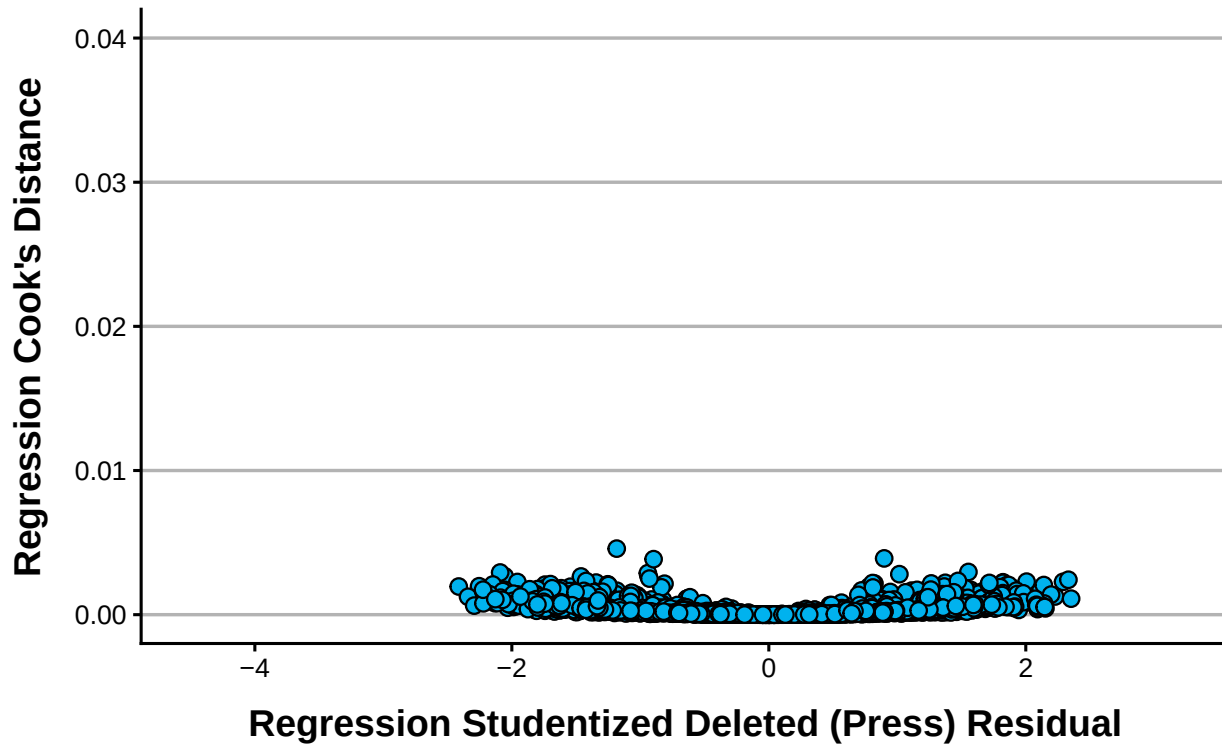
Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



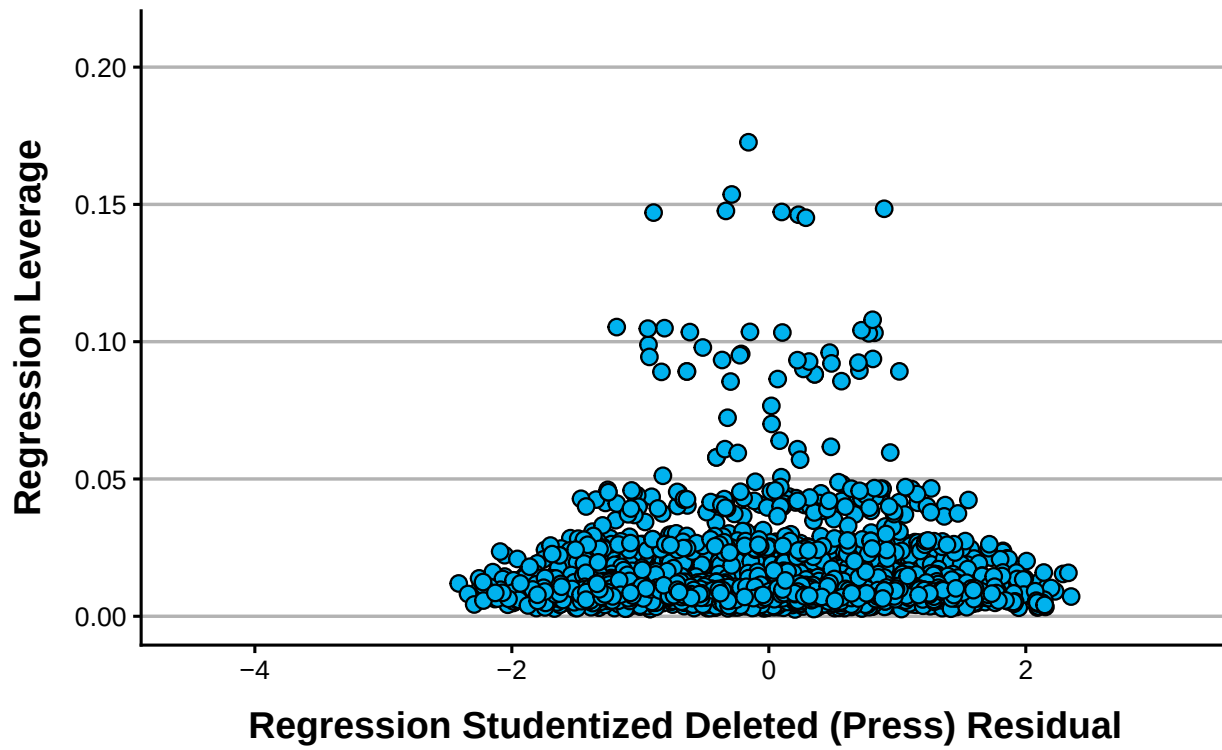
Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Model IV

Table 12: District 7 LN_TASP Model I assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2667	1.085437	1.216642	0.9992245	1.217586	39.54243	-0.847816

Table 13: District 7 LN_TASP Model II assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2667	1.082763	1.215931	0.9991601	1.216953	39.59413	-0.8443798

Table 14: District 7 LN_TASP Model I fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2490	2456	36	0.3874	0.3941	0.386	0.3848	0.3198	48.4173	0	2379.687	2583.388

Table 15: District 7 LN_TASP Model I coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.4931	279.0119	0.0412	0.0000	11.4124	11.5739
ln_landratio	0.2473	4.7071	0.0525	0.0000	0.1443	0.3503
ln_base_bldgsize_ratio	0.5030	8.8136	0.0571	0.0000	0.3911	0.6150
single_family1story	0.0456	1.7031	0.0268	0.0887	-0.0069	0.0982
smallest	-0.0570	-1.9379	0.0294	0.0527	-0.1146	0.0007
fair	-0.1254	-6.4135	0.0196	0.0000	-0.1638	-0.0871
poor	-0.2805	-5.4153	0.0518	0.0000	-0.3820	-0.1789
qbelow_avg	-0.0793	-3.8651	0.0205	0.0001	-0.1196	-0.0391
yb_1946_to_1960	0.0442	2.4051	0.0184	0.0162	0.0082	0.0803
yb_post_1961	0.2147	3.6995	0.0580	0.0002	0.1009	0.3286
crawl	0.2057	2.7572	0.0746	0.0059	0.0594	0.3521
ln_garagespaces	0.1350	6.6050	0.0204	0.0000	0.0949	0.1751
fireplace	0.0741	3.3628	0.0220	0.0008	0.0309	0.1173
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	-0.3168	-2.7838	0.1138	0.0054	-0.5400	-0.0937
central_air	0.0848	3.8708	0.0219	0.0001	0.0418	0.1277
ecf_7R703	-0.6267	-9.8404	0.0637	0.0000	-0.7516	-0.5018
ecf_7R704	-0.4076	-11.8288	0.0345	0.0000	-0.4752	-0.3401
ecf_7R705	-0.4115	-13.7675	0.0299	0.0000	-0.4701	-0.3529
ecf_7R706	-0.3882	-4.9894	0.0778	0.0000	-0.5407	-0.2356
ecf_7R707	-0.4164	-5.0606	0.0823	0.0000	-0.5778	-0.2551
ecf_7R708	-0.4943	-10.9669	0.0451	0.0000	-0.5827	-0.4059
ecf_7R709	-0.4670	-10.7884	0.0433	0.0000	-0.5519	-0.3822
ecf_7R710	-0.2886	-6.2972	0.0458	0.0000	-0.3784	-0.1987
ecf_7R711	-0.5175	-12.0326	0.0430	0.0000	-0.6018	-0.4331
ecf_7R712	-0.3507	-5.5119	0.0636	0.0000	-0.4755	-0.2260
ecf_7R713	-0.4628	-8.1592	0.0567	0.0000	-0.5740	-0.3516
ecf_7R714	NA	NA	NA	NA	NA	NA
ecf_7R715	-0.6155	-5.3102	0.1159	0.0000	-0.8427	-0.3882
ecf_7R716	-0.9272	-15.0618	0.0616	0.0000	-1.0479	-0.8065
ecf_7R717	-0.9735	-16.3122	0.0597	0.0000	-1.0905	-0.8565
ecf_7R718	-0.7420	-16.3311	0.0454	0.0000	-0.8311	-0.6529
ecf_7R719	-0.3275	-9.5059	0.0344	0.0000	-0.3950	-0.2599

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_7R720	-0.7199	-4.7918	0.1502	0.0000	-1.0145	-0.4253
ecf_7R722	-0.4924	-3.8607	0.1275	0.0001	-0.7425	-0.2423
ecf_7R724	-0.8467	-10.4910	0.0807	0.0000	-1.0050	-0.6885

Table 16: District 7 LN_TASP Model II fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2490	2457	35	0.3878	0.3927	0.3847	0.3852	0.3195	49.6394	0	2383.81	2581.691

Table 17: District 7 LN_TASP Model II coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	9.6351	53.5846	0.1798	0.0000	9.2825	9.9876
ln_landratio	0.2605	4.9758	0.0524	0.0000	0.1578	0.3632
lnsbldsfint	0.2930	10.8246	0.0271	0.0000	0.2399	0.3461
two_plus_baths	0.1282	1.8968	0.0676	0.0580	-0.0043	0.2606
fair	-0.1298	-6.6453	0.0195	0.0000	-0.1680	-0.0915
poor	-0.2916	-5.6277	0.0518	0.0000	-0.3931	-0.1900
qbelow_avg	-0.0800	-3.9285	0.0204	0.0001	-0.1199	-0.0401
yb_1946_to_1960	0.0503	2.7847	0.0181	0.0054	0.0149	0.0857
yb_post_1961	0.2308	3.9909	0.0578	0.0001	0.1174	0.3441
crawl	0.1987	2.6634	0.0746	0.0078	0.0524	0.3449
garage	0.1128	6.4177	0.0176	0.0000	0.0783	0.1472
fireplace	0.0674	3.0664	0.0220	0.0022	0.0243	0.1106
elec_baseboard	NA	NA	NA	NA	NA	NA
wall	-0.3028	-2.6566	0.1140	0.0079	-0.5264	-0.0793
central_air	0.0878	4.0125	0.0219	0.0001	0.0449	0.1308
ecf_7R703	-0.6153	-9.6574	0.0637	0.0000	-0.7402	-0.4903
ecf_7R704	-0.4001	-11.6046	0.0345	0.0000	-0.4678	-0.3325
ecf_7R705	-0.4048	-13.4552	0.0301	0.0000	-0.4638	-0.3458
ecf_7R706	-0.3851	-4.9344	0.0781	0.0000	-0.5382	-0.2321
ecf_7R707	-0.4014	-4.8689	0.0824	0.0000	-0.5631	-0.2397
ecf_7R708	-0.4845	-10.6972	0.0453	0.0000	-0.5734	-0.3957
ecf_7R709	-0.4611	-10.6097	0.0435	0.0000	-0.5463	-0.3759
ecf_7R710	-0.2818	-6.1107	0.0461	0.0000	-0.3723	-0.1914
ecf_7R711	-0.5093	-11.7773	0.0432	0.0000	-0.5941	-0.4245
ecf_7R712	-0.3385	-5.3120	0.0637	0.0000	-0.4635	-0.2136
ecf_7R713	-0.4593	-8.0779	0.0569	0.0000	-0.5707	-0.3478
ecf_7R714	NA	NA	NA	NA	NA	NA
ecf_7R715	-0.6197	-5.3345	0.1162	0.0000	-0.8475	-0.3919
ecf_7R716	-0.9203	-14.7884	0.0622	0.0000	-1.0424	-0.7983
ecf_7R717	-0.9625	-15.9121	0.0605	0.0000	-1.0811	-0.8439
ecf_7R718	-0.7342	-15.9928	0.0459	0.0000	-0.8242	-0.6442
ecf_7R719	-0.3149	-8.9090	0.0354	0.0000	-0.3843	-0.2456
ecf_7R720	-0.7048	-4.6894	0.1503	0.0000	-0.9995	-0.4101
ecf_7R722	-0.4769	-3.7204	0.1282	0.0002	-0.7282	-0.2255
ecf_7R724	-0.8354	-10.2993	0.0811	0.0000	-0.9944	-0.6763

